

Lab 13: Bayes Theorem

Aim

To implement **Bayes Theorem** to calculate the probability of an event based on given prior probabilities and conditional probabilities.

Algorithm

1. Define the prior probability of the event.
2. Define the conditional probabilities.
3. Calculate the total probability of the evidence.
4. Apply Bayes Theorem.
5. Display the posterior probability.

Program

```
# Lab 13: Bayes Theorem
```

```
# Given probabilities
```

```
P_Disease = 0.01
```

```
P_Positive_given_Disease = 0.99
```

```
P_Positive_given_No_Disease = 0.05
```

```
# Probability of not having the disease
```

```
P_No_Disease = 1 - P_Disease
```

```
# Total probability of testing positive
```

```
P_Positive = (
```

```
    P_Positive_given_Disease * P_Disease
```

```
    + P_Positive_given_No_Disease * P_No_Disease
```

)

Bayes Theorem

```
P_Disease_given_Positive = (  
    P_Positive_given_Disease * P_Disease  
) / P_Positive
```

```
print("P(Disease | Positive) =", P_Disease_given_Positive)
```

Sample Output

P(Disease | Positive) = 0.16666666666666669

Result

Thus, **Bayes Theorem was successfully implemented** to calculate the posterior probability of having a disease given that the test result is positive.